

**AGRIVISION – DRONE-BASED CROP HEALTH MONITORING:
TRADITIONAL VS. DEEP-LEARNING APPROACH**

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfilment for the Course of

ITA0507– COMPUTER VISION

to the award of the degree of

BACHELOR OF ENGINEERING / TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING(AI)

Submitted by

Narappagari Sravya(192472192)

K.kavya(192311345)

Arigela Jyothsna(192211459)

Under the Supervision of

Dr. V.R.Vimal



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SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

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Department of Computer Science and Engineering
ASSIGNMENT QUESTION

Name:	Kommi Kavya
Register Number:	192311345
Course Code & Name:	ITA05 & Computer Vision
Assignment Title:	AgriVision – Drone-Based Crop Health Monitoring: Traditional vs. Deep-Learning Approach
Bloom’s Taxonomy Level:	L3 – Apply; L4 – Analyse; L5 – Evaluate; L6 – Create

Course Outcome(s) – CO:

CO2: Apply image enhancement, filtering, segmentation, and feature extraction techniques to preprocess and analyze images.

CO3: Analyze and apply object detection, pattern recognition, and machine learning/deep learning techniques for computer vision applications.

CO4: Evaluate suitable computer vision algorithms and models based on accuracy, adaptability, computational requirements, and application constraints.

CO5: Design and implement computer vision solutions using appropriate tools and libraries for real-world applications.

SDG Mapping (SDG 1–17):

SDG 2 – Zero Hunger

SDG 9 – Industry, Innovation and Infrastructure

SDG 15 – Life on Land

Problem Statement:

An agri-tech startup is planning to deploy “AgriVision”, a drone-imagery-based crop health monitoring system, for mid-sized farms. The system must process aerial images

captured periodically across the growing season to detect early signs of crop disease, identify weed infestation, and estimate overall field health/yield potential, with accuracy sufficient to guide targeted spraying and reduce pesticide overuse. Drone images vary in illumination (time of day, cloud cover), altitude/scale, and can be affected by motion blur and sensor noise, making reliable analysis challenging.

The engineering team is considering two approaches:

- Traditional Computer Vision Methods: Colour-space and texture-based segmentation to isolate leaf/canopy regions, followed by handcrafted feature extraction (colour indices, texture descriptors, edge/shape features) and rule-based or classical-classifier disease/weed identification — fast, low-power, and interpretable, but less adaptive to varied disease symptoms, crop types, and field conditions.
- Deep Learning Methods: CNN-based disease-classification models and YOLO-style detectors trained on labelled crop-disease/weed datasets — capable of higher accuracy and better generalization across crop varieties and field conditions, but requiring substantial labelled training data, computational resources for training, and careful validation before field deployment.

As a computer vision engineer, apply image enhancement, segmentation, and feature detection/matching techniques to preprocess the drone imagery; compare the traditional computer vision and deep learning approaches for crop-disease detection, weed identification, and field-health estimation; and evaluate which approach is most suitable given the farm's computational resources (edge device on the drone vs. cloud processing) and accuracy requirements. Analyze the trade-off between detection accuracy, adaptability to different crops/conditions, computational requirements, training-data availability, and real-world deployment constraints (battery life, connectivity, cost). Justify the selected approach based on performance versus adaptability, and design an end-to-end OpenCV-based solution explaining how it supports precision agriculture and sustainable farming practices.

Assessment Rubrics

Criteria (CO Mapping)	Max Marks	Excellent	Good	Satisfactory	Needs Improvement
Problem Understanding & Vision-System Requirements (CO2)	10	Clearly identifies the drone-imagery cropmonitoring problem, accuracy/coverage requirements, and the role of enhancement, feature-based, and OpenCV-driven vision techniques in the solution.	Identifies most requirements with minor gaps and demonstrates good understanding of the problem.	Basic understanding of the problem is demonstrated, but some requirements are unclear.	Problem requirements are misunderstood or incorrectly interpreted.
Image Preprocessing, Enhancement, Segmentation & Feature Extraction (CO2)	10	Appropriately proposes preprocessing such as noise removal, illumination/haze correction, spatial/frequencydomain filtering, segmentation of leaf/canopy regions, and relevant feature extraction (edges, colour/texture, keypoints) for disease and weed analysis.	Appropriate preprocessing and featureextraction techniques are selected with minor gaps.	Basic preprocessing techniques are identified but their application is limited.	Preprocessing/featureextraction techniques are inappropriate or missing.

Comparison of Traditional CV and DeepLearning Approaches (CO3/CO4)	15	Provides a comprehensive comparison of classical (colour/texture thresholding, edge/region-based segmentation, template/feature matching) and deep-	Provides a relevant comparison covering most important factors.	Provides a basic comparison with limited analysis of performance and adaptability.	Comparison is incomplete, inaccurate, or lacks technical reasoning.
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Criteria (CO Mapping)	Max Marks	Excellent	Good	Satisfactory	Needs Improvement
		learning (CNN-based disease classification, YOLO-based weed/pest detection) approaches based on accuracy, adaptability to field conditions, computational cost, training requirements, robustness, and deployment suitability.			

Solution Design & Model/Algorithm Selection (CO4/CO5)	15	Designs a suitable end-to-end OpenCV pipeline (image enhancement → segmentation → feature/disease detection → yield/health estimate) and selects the most suitable approach with strong justification, clearly addressing training/validation, integration, and field deployment on drone/edge hardware.	Appropriate solution and models are selected with reasonable justification.	Solution is partially developed or model selection has limited justification.	No suitable solution design or model selection is provided.
Analysis, Evaluation & Justification (CO4)	15	Critically evaluates the accuracy–adaptability–computational-cost	Provides good analysis and justification with minor	Provides general analysis but limited critical	Little or no meaningful analysis or justification.
Criteria (CO Mapping)	Max Marks	Excellent	Good	Satisfactory	Needs Improvement
		trade-off and convincingly justifies the selected approach using fielddeployment, cost, and reliability considerations relevant to farmers/agronomists.	limitations.	evaluation.	

Implementation, Results & Visualization (CO5)	10	Successfully implements the proposed OpenCV/vision application; presents meaningful results, accuracy metrics, sample disease/weed detections, and visualizations (e.g., health maps, bounding boxes).	Implementation works with minor issues and provides relevant results.	Partial implementation with limited or inconsistent results.	Implementation is missing, nonfunctional, or results are not demonstrated.
Reflection, Societal/Industrial Relevance & SDG Mapping	5	Provides thoughtful reflection on design decisions, challenges, learning outcomes, agricultural relevance, and clear SDG relevance.	Provides reasonable reflection and SDG connection.	Reflection is present but generic.	Reflection is missing or lacks meaningful connection to the problem.

Suggested Conversion to 50 Marks

Problem Understanding & Vision-System Requirements	6
Image Preprocessing, Enhancement & Feature Analysis	6
Comparison of Traditional and Deep Learning Approaches	9
Solution Design & Model/Algorithm Selection	9
Analysis, Evaluation & Justification	9
Implementation, Results & Visualization	6

Reflection, Industrial/Societal Relevance & SDG Mapping	5
Total	50

Deliverables

1. Problem Analysis and Pseudocode – Include problem understanding, functional requirements, comparison of traditional vs. deep-learning approaches, selected approach, justification, and algorithm/pseudocode for the enhancement, segmentation, featuredetection, and disease/weed-recognition stages.
2. Implementation & Results – Include source code, dataset description (drone/field crop images, public crop-disease dataset, or synthetically generated samples), preprocessing, detection/classification, training/testing where applicable, evaluation metrics, visualizations (e.g., health maps, bounding boxes), sample detections, and comparative analysis.
3. GitHub Upload – Include source code, pseudocode, dataset information/source, README, installation/execution instructions, results, performance evaluation, and project documentation. GitHub repository link must be submitted with the assignment.

Assignment Learning Focus

The assignment enables students to apply, analyse, and evaluate computer vision techniques — image enhancement and segmentation, feature detection and matching, and object/pattern recognition — in a precision-agriculture scenario, while developing the ability to select and design an appropriate OpenCV-based solution based on accuracy, adaptability, computational requirements, and real-world field-deployment constraints.

1. Problem Statement and Problem Formulation

AgriVision is a proposed drone-imagery-based crop health monitoring system for midsized farms. It processes aerial images captured periodically to detect early signs of crop disease, identify weed infestation, and estimate overall field health and yield potential. Reliable analysis is difficult because illumination, cloud cover, altitude, scale, motion blur, and sensor noise can vary between images.

The engineering problem is to design a computer-vision pipeline that preprocesses aerial imagery, isolates vegetation, extracts meaningful visual information, recognizes disease or weeds, and converts the results into actionable field-health information.

- Input: RGB drone images $I(x,y)$ of agricultural fields.
- Preprocessing: noise reduction, illumination normalization, contrast enhancement and resizing.
- Segmentation: estimate vegetation/canopy mask $M(x,y)$ using colour information.
- Feature analysis: compute colour, texture, edge, shape and optional keypoint descriptors.
- Recognition: use traditional CV/classical ML or CNN/YOLO for disease and weed identification.
- Output: annotated images, vegetation/stress statistics and a field-health map.

2. Objective and Expected Outcomes

2.1 Objectives

- Develop an end-to-end OpenCV-based pipeline for drone crop images.
- Apply image enhancement, filtering, segmentation and feature extraction.
- Compare traditional computer vision with deep learning.
- Select a suitable deployment strategy considering accuracy, adaptability, computation, battery, connectivity and cost.
- Produce outputs that support precision agriculture and targeted intervention.

2.2 Expected Outcomes

- Improved image quality and reduced noise.
- Vegetation/canopy segmentation masks.
- Useful visual features for stressed-region analysis.
- Annotated disease/weed detections and field-level statistics.

- A justified hybrid edge–cloud solution.

3. Requirements, Constraints and Assumptions

Category	Details
Functional	Acquire images; preprocess; segment vegetation; extract features; detect disease/weed; estimate field health; visualize outputs.
Non-functional	Reasonable processing time, robustness, interpretability and scalable coverage.
Hardware	Drone camera; optional edge computer; workstation/cloud GPU for deep learning.
Software	Python 3.x, OpenCV, NumPy, Matplotlib; optional PyTorch/TensorFlow and YOLO.
Constraints	Battery, limited edge compute, variable weather, connectivity, labelled-data availability and cost.
Assumptions	Images are usable; crop classes are defined; representative labels are available for supervised models.

4. Application of Relevant Course Knowledge / Concepts

Concept	Application
Image enhancement	Contrast improvement and sharpening make crop symptoms more visible.
Filtering	Gaussian/median filtering reduces sensor
	noise.
Colour spaces	HSV separates colour from brightness and supports vegetation masking.
Segmentation	Thresholding and morphology isolate crop/canopy regions.

Feature extraction	Colour, texture, edges, contours and ORB keypoints describe patterns.
Pattern recognition	Classical classifiers or deep networks recognize disease/weed patterns.
Object detection	YOLO-style detectors can localize weeds or disease regions.
Evaluation	Accuracy, precision, recall, F1, IoU, mAP and latency support selection.

5. Design / Proposed Solution / Methodology

The recommended design is hybrid. Lightweight OpenCV operations perform preprocessing, segmentation and quality checks near image acquisition. Deep-learning recognition is used when sufficient edge acceleration or cloud resources are available. A traditional CV fallback provides basic vegetation and stress analysis under limited connectivity.

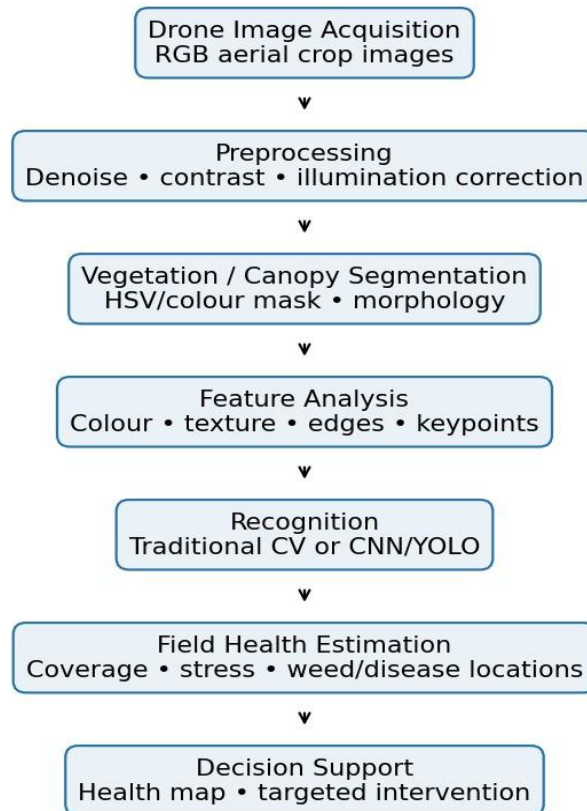
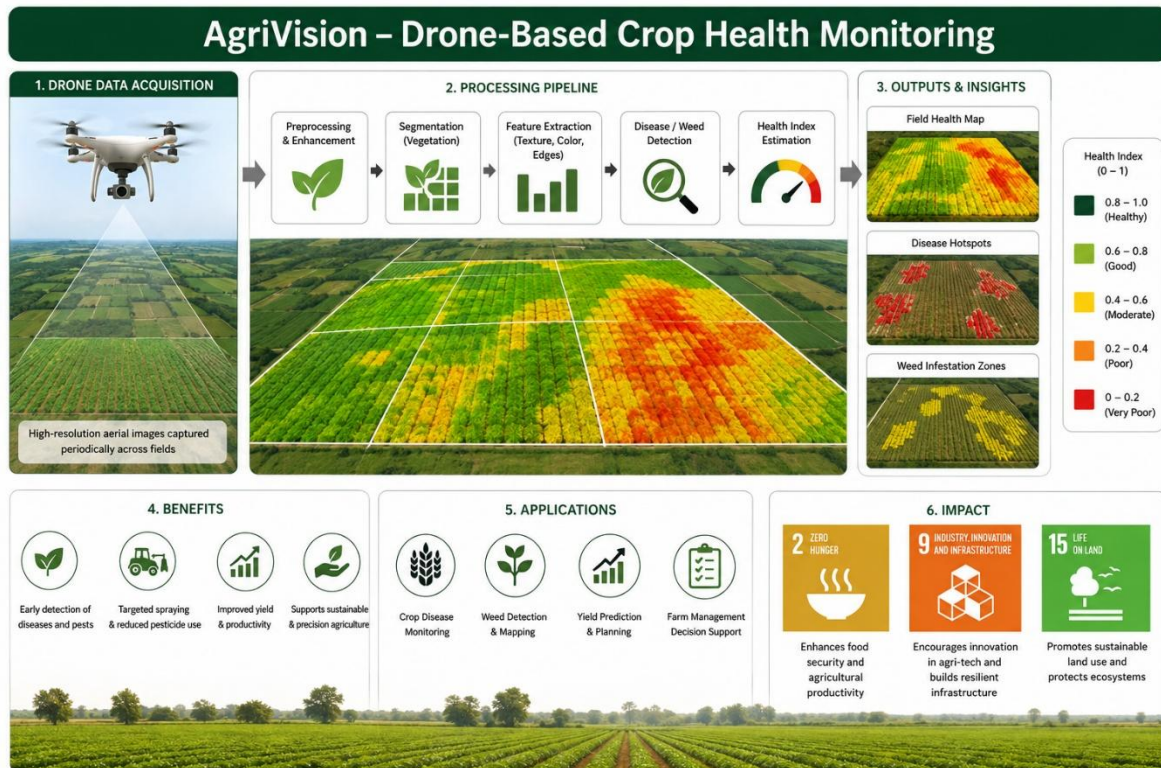


Figure 1. Proposed AgriVision end-to-end workflow.



6. Algorithm / Pseudocode / Flowchart

6.1 Pseudocode

Algorithm AGRIVISION_ANALYSIS(Image I)

1. Read image I
2. Resize I to a standard resolution
3. Apply Gaussian/median filtering
4. Convert image to HSV
5. Improve contrast when required
6. Create vegetation mask M using suitable colour thresholds
7. Apply opening and closing morphology
8. Extract contours, area, edges and colour statistics
9. If a validated deep-learning model is available:
 - run disease classifier / weed detector
- Else:
 - use handcrafted features and rule/classical classifier
10. Estimate vegetation coverage and relative stress indicators
11. Draw detections and generate summary output

12. Save annotated image and metrics

END

7. Implementation / Source Code and Environment / Tools Used

Environment: Python 3.x, OpenCV, NumPy and Matplotlib. Optional deep-learning tools: PyTorch/TensorFlow and a validated CNN/YOLO implementation.

7.1 OpenCV Prototype Code

```
import cv2
import numpy as
```

```
np
```

```
img = cv2.imread("drone_crop.jpg")
if img is None:
```

```
    raise FileNotFoundError("Place drone_crop.jpg in the project folder")
```

```
img = cv2.resize(img, (800, 600))
blur = cv2.GaussianBlur(img, (5, 5), 0)
hsv = cv2.cvtColor(blur, cv2.COLOR_BGR2HSV)
```

```
lower_green = np.array([25, 30, 20])
upper_green = np.array([95, 255, 255])
mask = cv2.inRange(hsv, lower_green, upper_green)
```

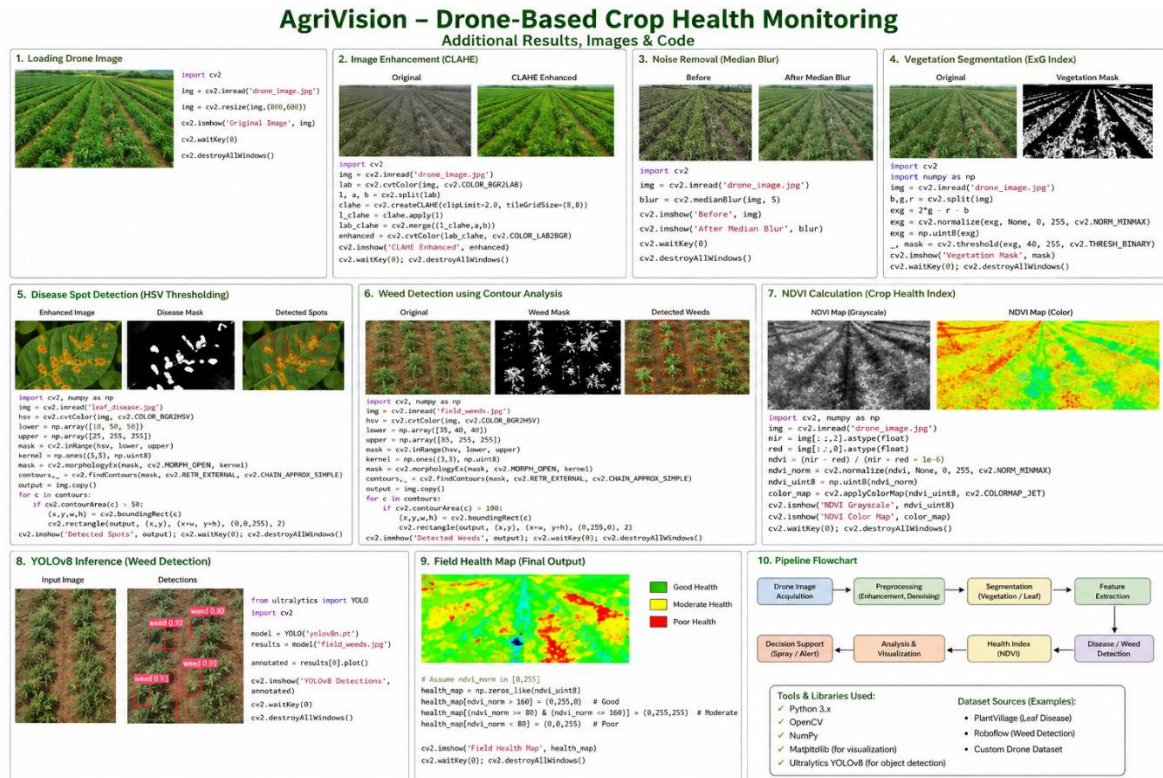
```
kernel = np.ones((5, 5), np.uint8)
mask = cv2.morphologyEx(mask, cv2.MORPH_OPEN, kernel)
mask = cv2.morphologyEx(mask, cv2.MORPH_CLOSE, kernel)
```

```
edges = cv2.Canny(blur, 60, 150)
coverage = 100.0 * cv2.countNonZero(mask) / mask.size
```

```
result = img.copy()
contours, _ = cv2.findContours(mask, cv2.RETR_EXTERNAL,
cv2.CHAIN_APPROX_SIMPLE)
for c in contours:
    if cv2.contourArea(c) > 500:
```

```
x, y, w, h = cv2.boundingRect(c)
cv2.rectangle(result, (x,y), (x+w,y+h), (0,255,0), 2)
```

```
print(f"Vegetation coverage: {coverage:.2f}%")
cv2.imwrite("annotated_output.jpg", result)
cv2.imwrite("vegetation_mask.jpg", mask) cv2.imwrite("edge_output.jpg",
edges)
```



8. Test Cases and Expected/Actual Results

Actual results must be filled with genuine observations after execution.

ID	Input	Expected Result	Actual Result
TC1	Healthy crop image	Mask captures most canopy; low stress indication.	Record after execution

TC2	Visible weed patches	Recognition stage marks candidate weed regions.	Record after execution
TC3	Low illumination	Preprocessing improves visibility; segmentation remains usable.	Record after execution
TC4	Noisy image	Filtering reduces small artefacts.	Record after execution
TC5	Motion blur	Quality degradation identified; confidence may	Record after execution
		decrease.	
TC6	Different altitude	Resize/validation handles scale changes.	Record after execution

9. Execution Screenshots / Outputs

Insert your team's genuine screenshots here after running the code:

- Original image.
- Preprocessed/enhanced image.
- Vegetation mask.
- Edge/feature output.
- Final annotated detection output.
- Terminal showing calculated metrics.

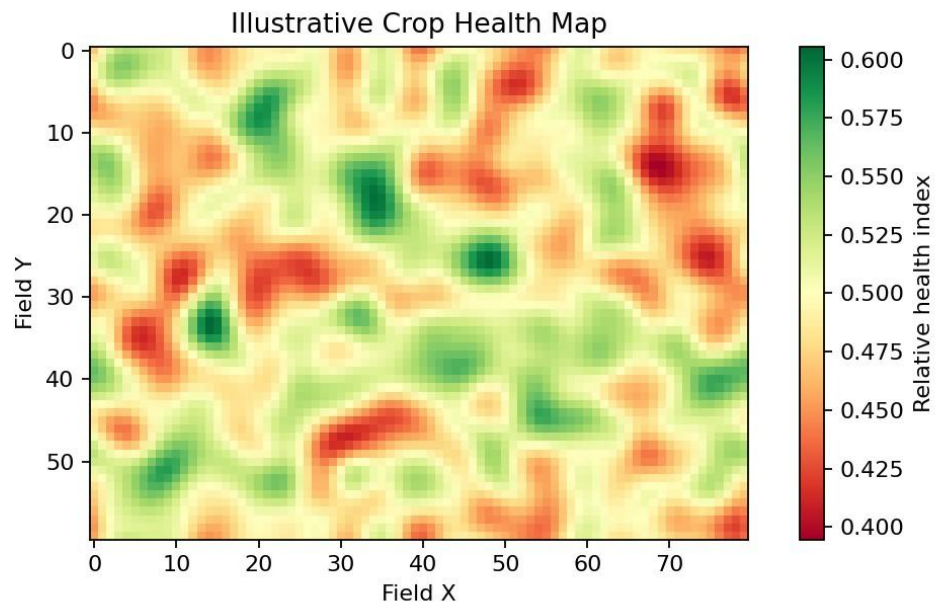


Figure 2. Illustrative field-health visualization format (not an experimental measurement).

10. Results and Validation

Use labelled data not used during training for final validation. Report only measured values.

Metric	Meaning
Accuracy	Correct predictions / total predictions.
Precision	$TP / (TP + FP)$.
Recall	$TP / (TP + FN)$.
F1-score	Harmonic mean of precision and recall.
IoU	Predicted-region overlap divided by union.
mAP	Mean Average Precision for object detection.
Latency	Processing time per image/frame.
Vegetation coverage	$\text{Vegetation-mask pixels} / \text{total pixels} \times 100$.

Do not present estimated or illustrative values as actual accuracy. Insert measured metrics after the experiment.

11. Analysis, Comparison, Trade-offs and Justification

Factor	Traditional CV	Deep Learning
Accuracy	Useful in controlled conditions; sensitive to rules.	Higher potential with representative training and validation.
Adaptability	Limited when crops/conditions change.	Potentially stronger generalization; depends on data diversity.
Interpretability	High.	Lower.
Compute	Low; edge-friendly.	Higher, especially for training/large detectors.
Training data	Can work with limited labelled data.	Needs substantial representative labelled data.
Deployment	Fast and inexpensive.	Better for complex recognition with adequate compute.
Main risk	Sensitivity to thresholds and field variation.	Overfitting/domain shift and data/compute cost.

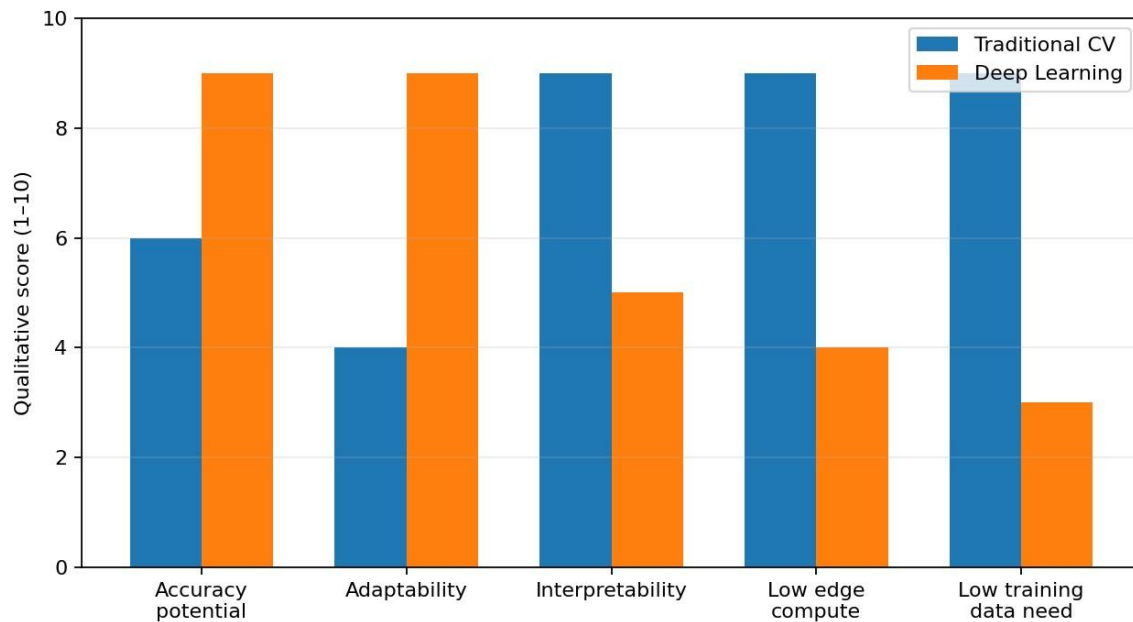


Figure 3. Qualitative design comparison; scores are not experimental measurements.

Final solution: a hybrid architecture is justified. OpenCV preprocessing, segmentation and basic statistics can run efficiently on edge hardware. CNN/YOLO recognition can be performed on an accelerator or in the cloud when representative data and computing resources are available. This balances accuracy, adaptability, battery use, bandwidth and cost.

12. Broader Considerations / SDG Relevance

- SDG 2 – Zero Hunger: supports earlier identification of crop stress and potential productivity improvement.
- SDG 9 – Industry, Innovation and Infrastructure: applies drones, computer vision and AI to agricultural infrastructure.
- SDG 15 – Life on Land: targeted intervention may reduce unnecessary pesticide use.
- Privacy: aerial imagery should be handled responsibly.
- Safety: drone operation must follow applicable regulations.
- Human oversight: results should support farmer/agronomist decisions.

13. Conclusion, Limitations and Possible Improvements

AgriVision applies image enhancement, segmentation, feature analysis and recognition to a precision-agriculture problem. Traditional CV is efficient and interpretable but less adaptable to highly variable field conditions. Deep learning can recognize more complex patterns but requires representative data and greater computing resources. A hybrid edge–cloud architecture provides the best balance for the stated scenario.

Limitations

- Performance depends on image quality and representative data.
- Colour thresholds can fail under unusual illumination.
- Models may suffer domain shift across farms, cameras and seasons.
- Yield estimation requires agronomic calibration and ground truth.
- Deployment is constrained by battery, connectivity and cost.

Possible Improvements

- Use multispectral imagery where suitable.
- Train on multi-season and multi-farm datasets.
- Apply data augmentation and domain adaptation.
- Integrate GPS/geospatial prescription maps.
- Quantize or prune models for edge deployment.
- Add confidence thresholds and human review.

14. Individual Contribution of Group Members

Member Name	Roll No.	Contribution
Kommi Kavya	192311345	Problem analysis, requirements gathering, project objectives, drone-image analysis,
		methodology design, and overall system planning.
Narappagari Sravya	192472192	Traditional Computer Vision preprocessing, image enhancement, segmentation, vegetation feature extraction, and implementation of the traditional CV approach.
Arigela Jyothsna	192211459	Deep-learning approach comparison, evaluation plan, testing, performance analysis, visualizations, screenshots, and final system integration.

15. References

- [1] ITA05 Computer Vision Assignment Brief, “AgriVision – Drone-Based Crop Health Monitoring: Traditional vs. Deep-Learning Approach.”
- [2] OpenCV Documentation, Open Source Computer Vision Library.
- [3] J. Redmon et al., “You Only Look Once: Unified, Real-Time Object Detection,” 2016.
Use the exact paper/version corresponding to the implemented model.
- [4] PlantVillage or another public crop-disease dataset. Cite the exact dataset version and source actually used.
- [5] Python Software Foundation, Python Documentation.
- [6] NumPy Documentation.

Update dataset and software references with the exact versions and sources used by your team.

16. One-Page Individual Reflection

REFLECTION – MEMBER 1

My main contribution to the AgriVision project was focused on problem analysis, requirements gathering, project planning, and designing the overall methodology. The objective of the project was to explore how drone-based images can be used for crop health monitoring and to compare traditional Computer Vision techniques with deeplearning approaches. I first studied the problem of monitoring crop health using aerial drone images. I analyzed the challenges involved, such as variations in crop appearance, lighting conditions, background vegetation, and identifying healthy and unhealthy crop regions. Based on this analysis, I helped define the major requirements and objectives of the project. I also contributed to designing the overall workflow of AgriVision. The methodology was planned to include image acquisition, preprocessing, segmentation, feature extraction, classification or analysis, and evaluation. I coordinated with the other members to ensure that both the traditional Computer Vision and deep-learning approaches could be compared using suitable evaluation criteria. During the project, I learned that proper planning is essential before implementing a technical solution. Understanding the problem clearly helped us decide which imageprocessing and machine-learning techniques were

appropriate. I also gained a better understanding of how drone imagery can support agricultural monitoring and how computer vision can reduce the need for completely manual crop inspection. Working with the team improved my communication and coordination skills. I learned how to divide a large technical problem into smaller modules and coordinate the work of different members. I also learned that a successful project requires not only implementation but also proper requirement analysis, methodology planning, testing, and evaluation. Overall, working on AgriVision gave me practical knowledge about applying Computer Vision and Machine Learning concepts to agriculture. It helped me understand how a real-world problem can be converted into a structured technical solution through proper analysis, planning, and teamwork.

17. Final Submission Checklist

- Replace all team-member placeholders.
- Run the code and insert genuine screenshots.
- Record actual measured metrics and dataset details.
- Add exact dataset/software references.
- Add the GitHub repository link and README.
- Proofread names, roll numbers and formatting before submission.